Scheme Theory

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1 Sheaves:

1.1 Presheaves, Sheaves, and Stalks:

Exercise 2.2.E. Let $\mathcal{F}(U)$ be the set of maps $U \to S$ that are locally constant, i.e. for any point $p \in U$, there is an open neighborhood of p where the function is constant. Show that this is a sheaf. We denote this, the constant sheaf, by $\underline{S}$.

Solution. This is clearly a presheaf with the usual restriction on functions. Thus, we only need to check identity and gluability.
The identity axiom is easy to verify as functions are determined by their values at points. Thus, if two functions agree on an open cover, they are the same function.

For gluability, let $\{U_i\}$ be an open cover with locally constant functions $\{f_i\}$ that agree on overlaps. Let f be the (set-theoretic) function that agrees with this information. We need to show that f is locally constant.
Let $p \in X$. Then there exists i such that $p \in U_i$. Thus, there exists a neighborhood $W \subseteq U_i$ containing p such that f_i is constant on it. Then f restricted to this is constant. Thus, f is locally constant.

Exercise 2.2.F. Let Y be a topological space. To each open set U of X, associate the set of continuous maps $U \to Y$. Show that this forms a sheaf. (2.2.D(b) and 2.2.E are special cases.)

Solution. We define the sheaf $\mathcal{F}$ on X as follows:

1. $\mathcal{F}(U)$ is the set of continuous maps to Y
2. For $V \subseteq U$, $res_{U,V}(f) = f|_V$, the usual restriction.

This is clearly a presheaf via the usual properties of functions. We have to show the identity and gluability axioms.
For the identity axiom, note that a continuous map is determined by the map on underlying

sets. Thus, if two functions agree on an open cover, they agree on the underlying set, and thus constitute the same function.
For the gluability axiom, let $\{U_i\}$ be an open cover of X and $f_i \in \mathcal{F}(U_i)$ be continuous functions to Y such that $f_i|_{U_i \cap U_j} \equiv f_j|_{U_i \cap U_j}$. Let f be the (unique) function of sets that agrees with this information. We need to show f is continuous.
Let $V \subseteq Y$ be open. Let $U = f^{-1}(V)$. Then,

$$U = f^{-1}(V) = \bigcup_i f_i^{-1}(V)$$

with each of the $f_i^{-1}(V)$ open (pre-image of an open set under a continuous function). Let $p \in U$ be arbitrary. Then, $p \in f_i^{-1}(V)$ for some i. Thus, U is open. Thus, f is continuous.

Exercise 2.2.G.

1. Suppose we are given a continuous map $\mu : Y \to X$. Show that "section of μ" form a sheaf. That is, to each open set U of X associate the set of continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}_U$. Show that this forms a sheaf.
2. Suppose that Y is a topological group. Show that continuous maps to Y form a sheaf of groups.

Solution. **TODO:**

Exercise 2.2.H. Suppose you have a map $\pi : X \to Y$ and a presheaf $\mathcal{F}$ on X. Construct the pushforward presheaf on Y by $\pi_*(V) := \mathcal{O}(\pi^{-1}(V))$ for $V \subseteq Y$. Prove that this is a presheaf and is a sheaf if $\mathcal{F}$ is.
Solution. **TODO:**

Exercise 2.2.I. Show that the pushforward of sheaves induces a map on stalks $\pi_*(\mathcal{F})_q \to \mathcal{F}_p$ where $q = \pi(p)$.
Solution. **TODO:**

Exercise 2.2.J. If $(X, \mathcal{O}_X)$ is a ringed space and $\mathcal{F}$ is a $\mathcal{O}_X$-module, show that $\mathcal{F}_p$ is a $\mathcal{O}_{X,p}$-module.
Solution. As $\mathcal{F}$ is a $\mathcal{O}_X$-module, we have the following commutative diagram for $V \subseteq U$:

$$\begin{tikzcd}
\mathcal{O}(U) \times \mathcal{F}(U) \arrow[r] \arrow[d, "res_{U,V} \times res_{U,V}"'] & \mathcal{F}(U) \arrow[d, "res_{U,V}"] \\
\mathcal{O}(V) \times \mathcal{F}(V) \arrow[r] & \mathcal{F}(V)
\end{tikzcd}$$

Let $[(f, U)] \in \mathcal{O}_{X,p}$ and $[(\chi, V)] \in \mathcal{F}_p$. Define the action of $\mathcal{O}_{X,p}$ by $[(f, U)] \cdot [(\chi, V)] = [(f \cdot \chi, U \cap V)]$. We need to show this is well-defined with respect to the equivalence relation on sections yielding the stalk.
Suppose $[(f, U)] = [(g, U')]$. Then, there exists $W \subseteq U \cap U'$ such that $f|_W \equiv g|_W$. This also holds for $W \cap V$. Thus, $[(f \cdot \chi, U \cap V)] = [(g \cdot \chi, U' \cap V)]$ as the representatives agree over $W \cap V \subset U \cap U' \cap V$. An analogous argument holds for being well-defined with respect to the second entry.

1.2 Morphisms:

Exercise 2.3.A. If $\phi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves, then ϕ induces a morphism of stalks $\phi_p : \mathcal{F}_p \to \mathcal{G}_p$.

Solution. Let $[(f, U)] \in \mathcal{F}_p$. We define the induced map $\phi_p([(f, U)]) = [(\phi(U)(f), U)] \in \mathcal{G}_p$. We need to show this is well-defined.
If $(f, U) \simeq (g, V)$, then there exists open $W \subset U \cap V$, $p \in W$ such that $f|_W = g|_W$. By the definition of sheaf morphisms, this implies that:

$$\phi(U)(f)|_W = \phi(V)(g)|_W$$

Thus, $[(\phi(U)(f), U)] = [(\phi(V)(g), V)]$.

Exercise 2.3.B. Suppose $\pi : X \to Y$ is continuous. Show that the pushforward gives a functor $\pi_* : Sets_X \to Sets_Y$.

Solution. Define the pushforward functor as follows:

1. For $\mathcal{F} \in Sets_X$, $\pi_*(\mathcal{F})(U) := \mathcal{F}(\pi^{-1}(U))$ for $U \subseteq Y$ open.
2. For $f : \mathcal{F} \to \mathcal{G}$ sheaf morphism with $\mathcal{F}, \mathcal{G} \in Sets_X$, define $\pi_*(f) : \pi_*(\mathcal{F}) \to \pi_*(\mathcal{G})$ by $\pi_*(f)(U) := f(\pi^{-1}(U))$ (this is a morphism from $\mathcal{F}(\pi^{-1}(U))) \to \mathcal{G}(\pi^{-1}(U))$.

We need to show that π_* preserves identities and compositions.
Let $\mathcal{F}$ be a sheaf on X and $\mathrm{id}_{\mathcal{F}}$ be the identity morphism. Then, we note that this preserves identities:

$$\pi_*(\mathrm{id}_{\mathcal{F}})(U) := \mathrm{id}_{\mathcal{F}}(\pi^{-1}(U)) = \mathrm{id}_{\mathcal{F}(\pi^{-1}(U))} = \mathrm{id}_{\pi_*(\mathcal{F})(U)}$$

Next, let $\mathcal{F}, \mathcal{G}, \mathcal{H}$ be sheafs on X with sheaf morphism $\phi : \mathcal{F} \to \mathcal{G}$ and $\psi : \mathcal{G} \to \mathcal{H}$. Let $f \in \pi_*(\mathcal{F})(U) = \mathcal{F}(\pi^{-1}(U))$ be a section. Then,

$$\pi_*(\phi)(f) = \phi(\pi^{-1}(U))(f) \in \mathcal{G}(\pi^{-1}(U)) = \pi_*(\mathcal{G})(U)$$

Similarly,

$$\pi_*(\psi)(\phi(\pi^{-1}(U))(f)) = \psi(\pi^{-1}(U))(\phi(\pi^{-1}(U))(f)) \in \mathcal{H}(\pi^{-1}(U)) = \pi_*(\mathcal{G})(U)$$

Overall,

$$\pi_*(\psi) \circ \pi_*(\phi)(f) := (\psi(\pi^{-1}(U)) \circ \phi(\pi^{-1}(U)))(f) = \pi_*(\psi \circ \phi)$$

Exercise 2.3.C. Suppose $\mathcal{F}$ and $\mathcal{G}$ are two sheaves of sets on X. (In fact, it will suffice that $\mathcal{F}$ is a presheaf.) Let $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ be the collection of data

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})(U) := \mathrm{Mor}(\mathcal{F}|_U, \mathcal{G}|_U)$$

Show that this is a sheaf of sets on X.

Solution. We define the restriction maps naturally as a sheaf morphism contains data for a sheaf morphism on a subspace. This is clearly a presheaf of sets. To prove this is a sheaf, we verify the axioms:

1. (Identity). Let $\{U_i\}$ be an open cover of U. Let $\phi, \psi \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U)$ be two sections such that $\phi|_{U_i} = \psi|_{U_i}$ for all i. Let $W \subseteq U$ be some open set. It suffices to show that $\phi(W) = \psi(W)$ as maps.
   Note that $\{W \cap U_i\}$ cover W. Moreover, we have $\phi(W \cap U_i) = \psi(W \cap U_i)$ (as $\phi|_{U_i} = \psi|_{U_i}$). Let $f \in \mathcal{F}(W)$. Then,
   $$\phi(W \cap U_i)(f|_{W \cap U_i}) = \psi(W \cap U_i)(f|_{W \cap U_i}) \in \mathcal{G}(W \cap U_i)$$
   As $\mathcal{G}$ is a sheaf, we get $\phi(W)(f) = \psi(W)(f)$ by the identity axiom. As f was arbitrary, this implies $\phi(W) = \psi(W)$. We are done.

2. (Gluability). Let $\{U_i\}$ be an open cover of U. Let $\phi_i \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U_i)$ such that $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$. We need to construct $\phi \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U)$ which restricts to these. Let $W \subseteq U$ be open and $f \in \mathcal{F}(W)$. Note that $\{W \cap U_i\}$ covers W. Moreover,
   $$\phi_i(W \cap U_i)(f|_{W \cap U_i}) \in \mathcal{G}(W \cap U_i)$$
   Next, note that: $\phi_i(W \cap U_i \cap U_j) = \phi_j(W \cap U_i \cap U_j)$. Thus, on $W \cap U_i \cap U_j$, we have:
   $$\phi_i(W \cap U_i \cap U_j)(f|_{W \cap U_i \cap U_j}) = \phi_j(W \cap U_i \cap U_j)(f|_{W \cap U_i \cap U_j})$$
   By gluability on $\mathcal{G}$, we get a section in $\mathcal{G}(U)$, call it g, such that $g|_{W \cap U_i} = \phi_i(W \cap U_i)(f|_{W \cap U_i})$. Let $\phi(W)(f) = g$. This defines $\phi \in \mathcal{H}om(\mathcal{F}, \mathcal{G})(U)$ as we have defined ϕ for every open $W \subseteq U$. Moreover, we note that this restricts appropriately as for $W \subseteq U_i$ our definition collapses to $\phi(W \cap U_i) = \phi_i(W \cap U_i) = \phi_i(W)$.

1.3 Abelian Category of Presheaves:

Exercise 2.3.E. Show that $\ker_{pre} \Phi$ is a presheaf. (Hint: if $V \hookrightarrow U$, define the restriction map by chasing the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & \ker_{pre} \Phi(U) \arrow[r] \arrow[d, dashed, "\exists!"] & \mathcal{F}(U) \arrow[r] \arrow[d, "res_{U,V}"] & \mathcal{G}(U) \arrow[d, "res_{U,V}"] \\
0 \arrow[r] & \ker_{pre} \Phi(V) \arrow[r] & \mathcal{F}(V) \arrow[r] & \mathcal{G}(V)
\end{tikzcd}$$

You should check that the restriction maps compose as desired.

Solution. Define the restriction map using the universal property of $\ker(\Phi(V))$. The uniqueness of this morphism implies that the restriction maps are compatible.

Exercise 2.3.F.

Solution. **TODO:**

Exercise 2.3.I.

Solution. **TODO:**

Exercise 2.3.J. Let $X = \mathbb{C}$ with the classical topology, let $\underline{\mathbb{Z}}$ be the constant sheaf on X associated to $\mathbb{Z}$, let $\mathcal{O}_X$ be the sheaf of holomorphic functions, and let $\mathcal{F}$ be the *presheaf* of functions admitting a holomorphic logarithm. Describe an exact sequence of presheaves on X:

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{F} \longrightarrow 0$$

where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion and $\mathcal{O}_X \to \mathcal{F}$ is given by

$$f \longmapsto \exp(2\pi i f)$$

(Be sure to verify exactness.) Show that $\mathcal{F}$ is *not* a sheaf. (*Hint:* $\mathcal{F}$ does not satisfy the gluing axiom. The problem is that there are functions that do not have a logarithm but locally have a logarithm.)

Solution. **TODO:**

1.4 Stalks:

Exercise 2.4.A. Prove that a section of a sheaf of sets is determined by its germs, i.e., the natural map

$$\mathcal{F}(U) \longrightarrow \prod_{p \in U} \mathcal{F}_p$$

is injective.

Solution. Let $f, g \in \mathcal{F}(U)$ be two sections whose germs agree at every point, i.e. for all $p \in U$, $[(f, U)] = [(g, U)]$. Then, there exists a $W_p \subseteq U$ such that $f|_{W_p} = g|_{W_p}$. Note that $\{W_p\}_{p \in U}$ form an open cover of U. By the identity axiom, we are done.

Exercise 2.4.B. Show that the support of section $\mathrm{Supp}(s)$ is closed in X.

Solution. We will show that the complement of the support is open. Let $p \in X \setminus \mathrm{Supp}(s)$. Then, $s_p = 0 \in \mathcal{F}_p$. This implies that there exists an open set W_p containing p such that $s|_{W_p} = 0|_{W_p}$, i.e. $W_p \subseteq X \setminus \mathrm{Supp}(s)$. Thus, $X \setminus \mathrm{Supp}(s)$ is open.

Exercise 2.4.C. Prove that any choice of compatible germs for a sheaf of sets $\mathcal{F}$ over U is the image of a section of $\mathcal{F}$ over U. (Hint: you will use gluability.)

Solution. Let $\{U_i\}$ be a cover of U and $f_i \in \mathcal{F}(U_i)$ be functions such that the germ of f_i at $p \in U_i$ is s_p. That is, s_p for $p \in U$ are compatible germs. First note that these sections agree on overlaps as they agree at each stalk in the intersection (2.4.A). Thus, by gluability, we are done.

Exercise 2.4.D. If ϕ_1 and ϕ_2 are morphisms from a presheaf of sets $\mathcal{F}$ to a sheaf of sets $\mathcal{G}$ that induce the same maps on each stalk, show that $\phi_1 = \phi_2$. Hint: consider the following diagram.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r] \arrow[d] & \mathcal{G}(U) \arrow[d] \\
\prod_{p \in U} \mathcal{F}_p \arrow[r] & \prod_{p \in U} \mathcal{G}_p
\end{tikzcd}$$

Solution. Assume ϕ_1 and ϕ_2 induce the same maps on stalks. Let $U \in X$ be an arbitrary open. It suffices to show that $\phi_1(U) = \phi_2(U)$. Let $f \in \mathcal{F}(U)$. Consider the map $\mathcal{F}(U) \to \prod_{p \in U} \mathcal{F}_p \to \prod_{p \in U} \mathcal{G}_p$. Since ϕ_1 and ϕ_2 induce the same maps on stalks, this map is identical regardless of whether the top morphism is ϕ_1 or ϕ_2. Thus, the germs of $\phi_1(U)(f)$ and $\phi_2(U)(f)$ agree for all $p \in U$. Then, as $\mathcal{G}$ is a sheaf, we use (2.4.A) to conclude $\phi_1(U)(f) = \phi_2(U)(f)$. As f was arbitrary, we are done.

Exercise 2.4.E. Show that a morphism of sheaves of sets is an isomorphism if and only if it induces an isomorphism of all stalks. Hint: Use 2.4.4.1. Once you have injectivity, show surjectivity, perhaps using Exercise 2.4.C, or gluability in some other way; this is more subtle.

Solution. **TODO:**

Exercise 2.4.F.

Solution. **TODO:**

1.5 Sheafification:

Exercise 2.4.H. Show that for any morphism of pre-sheaves $\phi : \mathcal{F} \to \mathcal{G}$ we have an induced map of sheaves $\phi^{sh} : \mathcal{F}^{sh} \to \mathcal{G}^{sh}$. Show that sheafification is a functor.

Solution. Consider the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\phi"] \arrow[d, "sh"'] & \mathcal{G} \arrow[d, "sh"] \\ \mathcal{F}^{sh} \arrow[r, dashed, "\phi^{sh}"'] & \mathcal{G}^{sh} \end{tikzcd}$$

where the map ϕ^{sh} is induced by the universal property of $\mathcal{G}^{sh}$. Functoriality is easy to see given this commutative diagram.

Exercise 2.4.I. Show that $\mathcal{F}^{sh}$ forms a sheaf.

Solution. Take the restriction map to be $(f_p)_{p\in U} \to (f_p)_{p\in V}$ for $V \subseteq U$. Note that the image also consists of compatible germs by taking appropriate intersections with V in the definition of a section.
To show identity, we let $\{U_i\}$ be an open cover of U with $f, g \in \mathcal{F}^{sh}(U)$ such that $f|_{U_i} = g|_{U_i}$ for all i. Then, $(f_p)_{p\in U_i} = (g_p)_{p\in U_i}$. Thus, $(f_p)_{p\in U} = (g_p)_{p\in U}$ as U_i cover. Thus, $f = g$.
To show gluability, let $f_i \in \{U_i\}$ be sections such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$. Note that for $p \in U$, we can pick a germ at p by taking $U_i \ni p$ and taking $(f_i)_p$. This is well-defined by the agreement on overlaps, i.e. $(f_i)_p = (f_j)_p$ for $p \in U_i \cap U_j$. These germs are compatible by construction and thus define a section in $\mathcal{F}(U)$.

Exercise 2.4.J. Describe a natural map of presheaves $\mathcal{F} \to \mathcal{F}^{sh}$.

Solution. Let $\phi(U) : \mathcal{F}(U) \to \mathcal{F}^{sh}(U)$ be $f \to (f_p)_{p\in U}$. This is a map of presheaves as it commutes with restrictions. That is, for $V \subseteq U$, we have $(f|_V)_p = f_p$ as they agree on $V \subseteq U$.

Exercise 2.4.K. Show that $\mathcal{F}^{sh}$ satisfies the universal property of sheafification.

Solution. Let $\phi : \mathcal{F} \to \mathcal{G}$ be a map of presheaves with $\mathcal{G}$ a sheaf. We first construct the map $\psi : \mathcal{F}^{sh} \to \mathcal{G}$.
By (2.3.A), ϕ induces a morphism on stalks, $\phi_p : \mathcal{F}_p \to \mathcal{G}_p$ with $\phi_p(f_p) = \phi(f)_p$. First, we claim that if $(f_p)_{p\in U}$ with $f_p \in \mathcal{F}_p$ are compatible germs, then $(\phi_p(f_p))_{p\in U}$ are also compatible. Clearly, if for all p there exists $V \subseteq U$ containing p and $s \in \mathcal{F}(V)$ such that $s_q = f_q$ for all $q \in V$. But then, $\phi(s)_q = \phi_q(s_q) = \phi_q(f_q)$ with $\phi(s) \in \mathcal{G}(V)$. Next, using (2.4.$A$)

and (2.4.C), we note that sections of a sheaf are uniquely determined by compatible germs. Thus, we have $g \in \mathcal{G}(U)$ with $\phi(f_p) = g_p$. Thus, we get a maps $\psi(U) : \mathcal{F}^{sh}(U) \to \mathcal{G}(U)$ by $(f_p)_{p \in U} \mapsto g$ with $g_p = \phi_p(f_p)$.

Clearly, ψ defines a map of sheaves as restriction commutes with ψ, i.e. for $V \subseteq U$, $g|_V$ satisfies $(g|_V)_p = \phi_p(f_p)$ for $p \in V$ and by (2.4.A) is unique. Thus, we only need to show $\phi = \psi \circ (\cdot)^{sh}$.

Let $f \in \mathcal{F}(U)$. Then,

$$\psi(U) \circ (\cdot)^{sh}(U)(f) = \psi(U)((f_p)_{p \in U}) = g$$

with $g_p = \phi_p(f_p)$ for all $p \in U$. We note that $\phi(U)(f)$ satisfies this property, so my uniqueness, $\phi(U)(f) = g$. Hence, we win.

Exercise 2.4.L.

Solution. **TODO:**

Exercise 2.4.M.

Solution. **TODO:**

1.6 Subsheaves and Quotient Sheaves:

Exercise 2.4.N.

Solution. **TODO:**

Exercise 2.4.O.

Solution. **TODO:**

Exercise 2.4.P.

Solution. **TODO:**

1.7 Sheaf on a base:

Exercise 2.5.A. Let $\{B_i\}$ is the base of a topology on X. You are given the information $\{\mathcal{F}(B_i), res_{B_i,B_j}\}$. Show that you can recover the entire sheaf from this.

Solution. **TODO:**

Exercise 2.5.B. Verify that $F(B) \to \mathcal{F}(B)$ is an isomorphism, likely by showing that it is injective and surjective (or else by describing the inverse map and verifying that it is indeed inverse). (Hint: elements of $\mathcal{F}(B)$ are determined by stalks, as are elements of $F(B)$.)

Solution. **TODO:**

Exercise 2.5.C. (Morphisms of Sheaves $\leftrightarrow$ morphisms of sheaves on a base) Suppose $\{B_i\}$ is a base for the topology on X. Suppose you are given a sheaf morphism $\mathcal{F} \to \mathcal{G}$ on the base, i.e., you have for $B_j \subseteq B_i$:

$$\begin{tikzcd}
\mathcal{F}(B_i) \arrow[r] \arrow[d, "res_{B_i,B_j}"'] & \mathcal{G}(B_i) \arrow[d, "res_{B_i,B_j}"] \\
\mathcal{F}(B_j) \arrow[r] & \mathcal{G}(B_j)
\end{tikzcd}$$

Show that:

1. Verify that a morphism of sheaves is determined by the induced morphism of sheaves on a base.
2. Show that morphism of sheaves on a base gives a morphism of induced sheaves.

Solution. **TODO:**

Exercise 2.5.D. Suppose $X = \bigcup U_i$ is an open cover of X and we have sheaves $\mathcal{F}_i$ on U_i along with isomorphism $\phi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \to \mathcal{F}_j|_{U_i \cap U_j}$ that agree on triple overlaps, i.e. $\phi_{jk} \circ \phi_{ij} = \phi_{ik}$ (cocycle condition). Show that these can be glued together into a sheaf $\mathcal{F}$ on X which is unique up to unique isomorphism with obvious restrictions.

Solution. **TODO:**

2 Schemes:

2.1 The set underlying an affine scheme:

Exercise 3.2.A

Solution. **TODO:**

Exercise 3.2.C

Solution. **TODO:**

Exercise 3.2.D If k is a field, show that $\operatorname{Spec} k[x]$ has infinitely many points.

Solution. Assume there are only finitely many prime ideals in $k[x]$, $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$. As $k[x]$ is a PID, let $p_1, \cdots, p_n$ be the generators of the ideals. Then, $p_1 \cdots p_n + 1$ has gcd 1 with every prime ideal, thus is not contained in any of them (else these ideals would be the whole ring). Thus, there is a maximal ideal that contains this that is not amongst our finite primes. Contradiction.

Exercise 3.2.F

Solution. **TODO:**

Exercise 3.2.G Show that if A is an integral domain that is a finite k-algebra, then A is a field.

Solution. Define $\phi_a : A \to A$ as multiplication by a, where $a \neq 0$. Then, ϕ_a is a k-linear map. Moreover, it is injective as A is an integral domain. Thus, it is an isomorphism. So there exists $y \in A$ such that $\phi_a(y) = ay = 1$.

Exercise 3.2.J Suppose A is a ring, and I an ideal of A. Let $\phi\colon A \to A/I$. Show that ϕ^{-1} gives an inclusion-preserving bijection between prime ideals of A/I and prime ideals of A containing I. Thus we can picture $\operatorname{Spec} A/I$ as a subset of $\operatorname{Spec} A$.

Solution. Let J be an ideal in A/I. We first show that $\phi^{-1}(J)$ is an ideal containing I.

First, note that $0 \in J$, thus, $I \in \phi^{-1}(J)$. Next, take $f, g \in \phi^{-1}(J)$. Then, $\phi(f), \phi(g) \in J$, thus $\phi(f) + \phi(g) = \phi(f + g) \in J$. Thus, $f + g \in \phi^{-1}(J)$. Similarly, take $a \in A$ and $f \in \phi^{-1}(J)$. Then, $\phi(af) \in J$ thus $af \in \phi^{-1}(J)$.

Next, we need to show that this is a bijection. For injectivity, note that since the quotient is surjective, two sets with the same pre-image are equal. Similarly, since ϕ is surjective, if J is an ideal of A, $\phi(J)$ is an ideal of A/I. **TODO:**

Exercise 3.2.K

Solution. **TODO:**

Exercise 3.2.L

Solution. **TODO:**

Exercise 3.2.M If $\phi : B \to A$ is a map of rings and $\mathfrak{p}$ is a prime ideal of A, show that $\phi^{-1}(\mathfrak{p})$ is a prime ideal of B.

Solution. Let $f, g \in \phi^{-1}(\mathfrak{p})$ and $b \in B$. First, note that $f + g \in \phi^{-1}(\mathfrak{p})$ as $\phi(f + g) = \phi(f) + \phi(g) \in \mathfrak{p}$. Similarly, $af \in \phi^{-1}(\mathfrak{p})$ as $\phi(af) = \phi(a)\phi(f) \in \mathfrak{p}$. Thus, $\phi^{-1}(\mathfrak{p})$ is an ideal. Then, assume $fg \in \phi^{-1}(\mathfrak{p})$. Then $\phi(fg) = \phi(f)\phi(g) \in \mathfrak{p}$. Thus, either $\phi(f)$ or $\phi(g)$ in $\mathfrak{p}$. Thus, $f \in \phi^{-1}(\mathfrak{p})$ or $g \in \phi^{-1}(\mathfrak{p})$.

Exercise 3.2.N

Solution. **TODO:**

Exercise 3.2.O

Solution. **TODO:**

Exercise 3.2.P

Solution. **TODO:**

Exercise 3.2.Q

Solution. **TODO:**

Exercise 3.2.S Show that $\mathfrak{R}(A)$ is the intersection of all prime ideals of A.

Solution. **TODO:**

2.2 Topology of affine schemes:

Exercise 3.4.D Show that $\sqrt{I}$ is an ideal. Show that $V(\sqrt{I}) = V(I)$.

Solution. Consider the map $R \to R/I$. This maps $\sqrt{I} \to \mathfrak{R}$. By (3.2.$R(b)$), $\mathfrak{R}$ is an ideal. Thus, its preimage is an ideal containing I. However, the preimage is exactly $\sqrt{I}$ by definition of a radical.
Moreover, as $I \subseteq \sqrt{I}$, $V(\sqrt{I}) \subseteq V(I)$. Next, let $\mathfrak{p} \in V(I)$ and $f \in \sqrt{I}$ with $f^n \in I$. Then, as $I \subseteq \mathfrak{p}$, $f^n \in \mathfrak{p}$. Thus, $f \in \mathfrak{p}$ by primality.

Exercise 3.4.E

Solution. **TODO:**

Exercise 3.4.F

Solution. **TODO:**

Exercise 3.4.H

Solution. **TODO:**

Exercise 3.4.I

Solution. **TODO:**

Exercise 3.4.J Suppose $I \subseteq B$ is an ideal. Show that f vanishes on $V(I)$ if and only if $f \in \sqrt{I}$.

Solution. f vanishes on $V(I)$ if and only if f is contained in every prime ideal that contains I if and only if f is contained in every prime ideal in B/I if and only if $f \in \mathfrak{R}(B/I)$ if and only if $f^n \in I$ if and only if $f \in \sqrt{I}$.

Exercise 3.4.K

Solution. **TODO:**

2.3 Base for the topology:

Exercise 3.5.A Show that the distinguished open sets form a base for the topology of $\operatorname{Spec} R$.

Solution. Let $U \subseteq \operatorname{Spec} R$ be an open set. Then, $\operatorname{Spec} R \setminus U$ is (Zariski) closed, thus can be written as $V(I)$ for some ideal I. We claim that $U = \bigcup_{f \in I} D(f)$.
Let $\mathfrak{p} \in U$. Then, there is $f \in I$ such that $f \notin \mathfrak{p}$. That is, $\mathfrak{p} \in D(f)$.
Conversely, let $\mathfrak{p} \in D(f)$ for $f \in I$. Then, $f \notin \mathfrak{p}$. Thus, $\mathfrak{p} \notin V(I)$. Thus, $\mathfrak{p} \in U$.
We are done.

Exercise 3.5.B Suppose $f_i \in A$ as i runs over some index set J. Show that

$$\bigcup_{i \in J} D(f_i) = \operatorname{Spec} A$$

if and only if $(\{f_i\}_{i \in J}) = A$, or equivalently and very usefully, if there are a_i $(i \in J)$, all but finitely many 0, such that

$$\sum_{i \in J} a_i f_i = 1.$$

Solution. Assume $(\{f_i\}_{i \in J}) = A$. Let $\mathfrak{p} \in \operatorname{Spec} A$ be a prime ideal. Then, there exists $i \in J$ such that $f_i \notin \mathfrak{p}$ as $\mathfrak{p} \subsetneq A$. Thus, $\mathfrak{p} \in D(f_i)$. Thus, $\operatorname{Spec} A = \bigcup_{i \in J} D(f_i)$.
Assume $\bigcup_{i \in J} D(f_i) = \operatorname{Spec} A$. Let $I = (\{f_i\}_{i \in J})$. If I was a proper ideal of A, then $I \subseteq \mathfrak{m}$, where $\mathfrak{m}$ is some maximal ideal. However, $\mathfrak{m} \in D(f_i)$ for some i, thus, $f_i \notin \mathfrak{m}$. Thus, $\mathfrak{m}$ cannot exist and I cannot be proper. Thus, $I = A$.
The latter fact follows from the definition of an ideal.

Exercise 3.5.C Show that if $\operatorname{Spec} A$ is an infinite union of distinguished open sets $\bigcup_{i \in J} D(f_i)$, then in fact it is a union of a finite number of these, i.e., there is a finite subset $J' \subset J$ so that

$$\operatorname{Spec} A = \bigcup_{j \in J'} D(f_j)$$

Solution. Assume $\operatorname{Spec} A = \bigcup_{i \in J} D(f_i)$. By 3.5.B, we know that $(\{f_i\}_{i \in J}) = A$. Thus,

$$1 = \sum_{i \in J' \subset J} a_i f_i$$

where $a_i \in A$ and J' is finite. Then, $(\{f_i\})_{i \in J'} = A$. By 3.5.B, we have $\operatorname{Spec} A = \bigcup_{i \in J'} D(f_i)$.

Exercise 3.5.D Show that

$$D(f) \cap D(g) = D(fg)$$

Solution. Let $\mathfrak{p} \in D(fg)$. Then, $fg \notin \mathfrak{p}$. Thus, $f, g \notin \mathfrak{p}$. Thus, $\mathfrak{p} \in D(f) \cap D(g)$.
Let $\mathfrak{p} \in D(f) \cap D(g)$. Then, $f, g \notin \mathfrak{p}$. As $\mathfrak{p}$ is a prime ideal, $fg \notin \mathfrak{p}$ (contrapositive of the prime property). Thus, $\mathfrak{p} \in D(fg)$. We are done.

Exercise 3.5.E Show that $D(f) \subseteq D(g)$ if and only if $f^n \in (g)$ for some $n \geq 1$ if and only if g is invertible in A_f.

Solution. First, we show (2) $\iff$ (3). Assume $f^n \in (g)$. Then, there exists $a \in A$ such that $f^n = ag$. Thus, in A_f, g has inverse $\frac{a}{f^n}$. Similarly, assume g is invertible in A_f. Then, there exists $\frac{a}{f^m} \in A_f$ with $a \in A$ satisfying $g\left(\frac{a}{f^n}\right) = 1$. Then, $ag = f^n$.
Next, we show (2) $\implies$ (1). Assume $f^n \in (g)$. Let $\mathfrak{p} \in V(g)$. Then, $f^n \in \mathfrak{p}$ and thus $f \in \mathfrak{p}$. This implies $V(g) \subseteq V(f)$, or, taking the complement $D(f) \subseteq D(g)$.
Finally, we show (1) $\implies$ (3). Assume $D(f) \subseteq D(g)$. Then, $V(g) \subseteq V(f)$. Thus, f vanishes on $V(g)$. By 3.4.J, $f \in \sqrt{(g)}$, i.e. $f^n \in (g)$ for some n.

Exercise 3.5.F Show that $D(f) = \emptyset$ if and only if $f \in \mathfrak{R}$.

Solution. This follows from $\mathfrak{R}$ being the intersection of all prime ideals. $f \in \mathfrak{R}$ if and only if it is contained in every prime ideal if and only if $D(f) = \emptyset$.

2.4 The dictionary, so far:

Exercise 3.7.A

Solution. **TODO:**

Exercise 3.7.B

Solution. **TODO:**

Exercise 3.7.C

Solution. **TODO:**

Exercise 3.7.D

Solution. **TODO:**

Exercise 3.7.E

Solution. **TODO:**

Exercise 3.7.F

Solution. **TODO:**

Exercise 3.7.G

Solution. **TODO:**

2.5 The Structure Sheaf:

Exercise 4.1.A Show that the natural map $A_f \to \mathcal{O}_{\operatorname{Spec} A}(D(f))$ is an isomorphism.

Solution. Note that $\mathcal{O}_{\operatorname{Spec} A}(D(f))$ is the localization of A at all functions g such that $D(f) \subseteq D(g)$. From (3.5.E), we know such g are already invertible in A_f. Certainly, $A_f \subseteq \mathcal{O}_{\operatorname{Spec} A}(D(f))$ (via appropriate further localization). Thus, $A_f \simeq \mathcal{O}_{\operatorname{Spec} A}(D(f))$.

Exercise 4.1.D Suppose M is an A module. Show that the following describes a sheaf $\widetilde{M}$ on the distinguished base. $\widetilde{M}(D(f))$ is the localization of M at the multiplicative set of all functions that do not vanish outside of $V(f)$. Restriction maps are obtained by further localizations. Also show that this is an $\mathcal{O}_{\operatorname{Spec} A}$ module.

Solution. **TODO:**

Exercise 4.1.E

Solution. **TODO:**

Exercise 4.1.F

Solution. **TODO:**

Exercise 4.1.G

Solution. **TODO:**

2.6 Schemes in general:

Exercise 4.3.A

Solution. **TODO:**

Exercise 4.3.B Suppose $f \in A$. Show that under the identification of $D(f)$ in $\operatorname{Spec} A$ with $\operatorname{Spec} A_f$ there is a natural isomorphism of ringed spaces

$$\big(D(f), \mathcal{O}_{\operatorname{Spec} A}|_{D(f)}\big) \;\cong\; \big(\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}\big).$$

Solution. **TODO:**

Exercise 4.3.C

Solution. **TODO:**

Exercise 4.3.D

Solution. **TODO:**

Exercise 4.3.E

Solution. **TODO:**

Exercise 4.3.F

Solution. **TODO:**

Exercise 4.3.G

Solution. **TODO:**

2.7 First examples:

Exercise 4.4.A

Solution. **TODO:**

Exercise 4.4.B

Solution. **TODO:**

Exercise 4.4.C

Solution. **TODO:**

Exercise 4.4.D

Solution. **TODO:**

Exercise 4.4.E

Solution. **TODO:**

2.8 The Proj construction:

Exercise 4.5.F

1. Show that an ideal I is homogenous if and only if it contains the degree n piece of each ofi ts elements.
2. Show that the set of homogenous ideals of a given $\mathbb{Z}$-graded ring $S_\bullet$ is closed under sum, product, intersection, and radical.

3. Show that a homogenous ideal $I \subseteq S_\bullet$ is prime if $I \neq S_\bullet$ and if for any homogenous $a, b \in S_\bullet$, if $ab \in I$ then either $a \in I$ or $b \in I$.

Solution. **TODO:**

Exercise 4.5.H Suppose $f \in S_+$ is homogenous. Give a bijection between the prime ideals of $((S_\bullet)_f)_0$ and the homogenous prime ideals of $(S_\bullet)_f$.

Solution. **TODO:**

Exercise 4.5.I Interpret the set of prime ideals of $((S_\bullet)_f)_0$ in terms of Proj $S_\bullet$. Show that this corresponds to $D_+(f)$.

Solution. **TODO:**

Exercise 4.5.K

Solution. **TODO:**

Exercise 4.5.L

Solution. **TODO:**

Exercise 4.5.M

Solution. **TODO:**

Exercise 4.5.N

Solution. **TODO:**

Exercise 4.5.O

Solution. **TODO:**

Exercise 4.5.Q

Solution. **TODO:**

Exercise 4.5.S

Solution. **TODO:**

3 First Properties of Schemes:

3.1 Topological and Noetherian Properties:

Exercise 3.6.I Show that the closed points of $\operatorname{Spec} A$ correspond to maximal ideals of A.

Solution. Let $\mathfrak{p} \in \operatorname{Spec} A$ be point. Then, if $\mathfrak{p}$ is contained in some maximal ideal $\mathfrak{m}$. Thus, $V(\mathfrak{p})$ contains $\mathfrak{m}$. Then, it is easy to see that $\{\mathfrak{p}\}$ closed forces $\mathfrak{p} = \mathfrak{m}$, i.e. $\mathfrak{p}$ is maximal.

Exercise 3.6.J

Solution. **TODO:**

Exercise 3.6.K

Solution. **TODO:**

Exercise 3.6.O

Solution. **TODO:**

Exercise 3.6.P

Solution. **TODO:**

Exercise 3.6.Q

Solution. **TODO:**

Exercise 3.6.S

Solution. **TODO:**

Exercise 3.6.T

Solution. **TODO:**

Exercise 3.6.U

Solution. **TODO:**

Exercise 3.6.V

Solution. **TODO:**

Exercise 3.6.W

Solution. **TODO:**

Exercise 3.6.X

Solution. **TODO:**

Exercise 3.6.Y

Solution. **TODO:**

Exercise 5.1.C Prove that if X is a scheme that has a finite cover $X = \bigcup \operatorname{Spec} A_i$ where A_i is Noetherian, then X is a Noetherian topological space.

Solution. **TODO:**

Exercise 5.1.E Show that if X is a quasicompact scheme, then every point has a closed point in its closure. That is, every non-empty closed subset of X contains a closed point of X.

Solution. Let Z be some non-empty irreducible closed subset of X. Let $U_1, \cdots, U_n$ be a finite affine open cover of X. Then, we have $Z = \bigcup_{i=1}^{n}(Z \cap U_i)$, all open in Z. As Z is irreducible, we must have $Z \cap U_i$ dense in Z for some i. Moreover, $Z \cap U_i$ is a non-empty closed subset of the affine scheme U_i, so it contains a closed point.

Exercise 5.1.F

Solution. **TODO:**

Exercise 5.1.G

Solution. **TODO:**

Exercise 5.1.H

Solution. **TODO:**

Exercise 5.1.I

Solution. **TODO:**

Exercise 5.1.J

Solution. **TODO:**

Exercise 5.1.K

Solution. **TODO:**

3.2 Reducedness and Integrality:

Exercise 5.2.B

Solution. **TODO:**

Exercise 5.2.C

Solution. **TODO:**

Exercise 5.2.D

Solution. **TODO:**

Exercise 5.2.E

Solution. **TODO:**

Exercise 5.2.F

Solution. **TODO:**

Exercise 5.2.G

Solution. **TODO:**

Exercise 5.2.H

Solution. **TODO:**

Exercise 5.2.I

Solution. **TODO:**

Exercise 5.2.J

Solution. **TODO:**

3.3 The Affine Communication Lemma:

Exercise 5.3.A

Solution. **TODO:**

Exercise 5.3.B

Solution. **TODO:**

Exercise 5.3.C

Solution. **TODO:**

Exercise 5.3.D

Solution. **TODO:**

Exercise 5.3.E

Solution. **TODO:**

Exercise 5.3.F

Solution. **TODO:**

Exercise 5.3.G

Solution. **TODO:**

Exercise 5.3.H

Solution. **TODO:**

3.4 Normality and Factoriality:

Exercise 5.4.A

Solution. **TODO:**

Exercise 5.4.B

Solution. **TODO:**

Exercise 5.4.C

Solution. **TODO:**

Exercise 5.4.D

Solution. **TODO:**

Exercise 5.4.E

Solution. **TODO:**

Exercise 5.4.F

Solution. **TODO:**

Exercise 5.4.G

Solution. **TODO:**

Exercise 5.4.H

Solution. **TODO:**

Exercise 5.4.I

Solution. **TODO:**

Exercise 5.4.J

Solution. **TODO:**

Exercise 5.4.K

Solution. **TODO:**

Exercise 5.4.L

Solution. **TODO:**

Exercise 5.4.M

Solution. **TODO:**

Exercise 5.4.N

Solution. **TODO:**